

Week-7 Practical

1. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
priyanka@DESKTOP-HOISO7: ~$ nano src/memory_addresses.c *
GNU nano 8.7.1 src/memory_addresses.c *
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;
static int static_var = 20;

void code_function() {
    printf("Code address : %p\n", (void *)code_function);
}

int main() {
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    printf("Memory Addresses\n");
    printf("-----\n");

    code_function();

    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address : %p\n", (void *)heap_var);
    printf("Stack address : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

Output:

```
priyanka@DESKTOP-HOISO7: ~$ cd ~/OS_Skill
mkdir -p Week-7/src Week-7/bin
cd Week-7
priyanka@DESKTOP-HOISO7: ~/OS_Skill/Week-7$ nano src/memory_addresses.c
priyanka@DESKTOP-HOISO7: ~/OS_Skill/Week-7$ gcc src/memory_addresses.c -o bin/memory_addresses
priyanka@DESKTOP-HOISO7: ~/OS_Skill/Week-7$ ./bin/memory_addresses
Memory Addresses
-----
Code address : 0x5f8dd46911c9
Global address : 0x5f8dd4694010
Static address : 0x5f8dd4694014
Heap address : 0x5f8de5084010
Stack address : 0x7ffc011a4c7c
```

2. Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
priyanka@DESKTOP-HOISO7I x + v - □ ×
GNU nano 8.7.1 src/memory_maps.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;

int main() {
    int stack_var = 200;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 300;

    printf("Process PID: %d\n", getpid());
    printf("Global address: %p\n", (void *)&global_var);
    printf("Stack address : %p\n", (void *)&stack_var);
    printf("Heap address  : %p\n", (void *)heap_var);

    printf("\nProcess is running...\n");
    printf("Use another terminal to inspect /proc/%d/maps\n", getpid());

    while (1) {
        sleep(1);
    }

    free(heap_var);

    return 0;
}
```

Output:

```
priyanka@DESKTOP-HOISO7I x + v - □ ×
priyanka@DESKTOP-HOISO7A:~$ cd ~/OS_Skill
mkdir -p Week-7/src Week-7/bin
cd Week-7
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ nano src/memory_addresses
.c
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ gcc src/memory_addresses.
c -o bin/memory_addresses
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ ./bin/memory_addresses
Memory Addresses
-----
Code address : 0x5f8dd46911c9
Global address : 0x5f8dd4694010
Static address : 0x5f8dd4694014
Heap address : 0x5f8de5084010
Stack address : 0x7ffc011a4c7c
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ nano src/memory_maps.c
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ gcc src/memory_maps.c -o
bin/memory_maps
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ ./bin/memory_maps
Process PID: 757
Global address: 0x5e29b432b010
Stack address : 0x7ffcd7f53ec
Heap address : 0x5e29b577c010

Process is running...
Use another terminal to inspect /proc/757/maps

priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ grep heap /proc/757/
maps
5e29b577c000-5e29b579d000 rw-p 00000000 00:00 0
[heap]
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ grep stack /proc/757
/maps
7ffcd7d6000-7ffcd7f7000 rw-p 00000000 00:00 0
[stack]
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-7$ pmap 757
757: ./bin/memory_maps
00005e29b4327000 4K r---- memory_maps
00005e29b4328000 4K r-x-- memory_maps
00005e29b4329000 4K r---- memory_maps
00005e29b432a000 4K r---- memory_maps
00005e29b432b000 4K rw--- memory_maps
00005e29b577c000 132K rw--- [ anon ]
000071b8ada00000 160K r---- libc.so.6
000071b8ada28000 1632K r-x-- libc.so.6
000071b8adb00000 312K r---- libc.so.6
000071b8adc0e000 16K r---- libc.so.6
000071b8adc12000 8K rw--- libc.so.6
000071b8adc14000 52K rw--- [ anon ]
000071b8adcb7000 12K rw--- [ anon ]
000071b8adcbf000 8K rw--- [ anon ]
000071b8adcc1000 16K r---- [ anon ]
000071b8adcc5000 8K r---- [ anon ]
000071b8adcc7000 8K r-x-- [ anon ]
000071b8adcc9000 4K r---- ld-linux-x86-64.so.2
000071b8adcca000 188K r-x-- ld-linux-x86-64.so.2
000071b8adccf000 44K r---- ld-linux-x86-64.so.2
000071b8add04000 8K r---- ld-linux-x86-64.so.2
000071b8add06000 4K rw--- ld-linux-x86-64.so.2
```